KCL AND KVL

CIRCUIT ANALYSIS METHOD

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TOPIC OUTLINE

Circuit Convention

Kirchhoff's Current Law (KCL)

Kirchhoff's Voltage Law (KVL)

CIRCUIT CONVENTION

CONVENTION

A **convention** is a widely accepted practice, method, or behavior that is followed by common **agreement** or tradition, rather than by formal rules.

Color coding in Offices

red – urgent documents

blue – general files

green – financial records

This is a common practice but not formally regulated.

STANDARD

A **standard** is a formal, established guideline, rule, or specification that is often **mandatory** and enforced by an authoritative body or organization.

IEC 60062 Resistor Color Code

black – 0

brown – 1

red – 2

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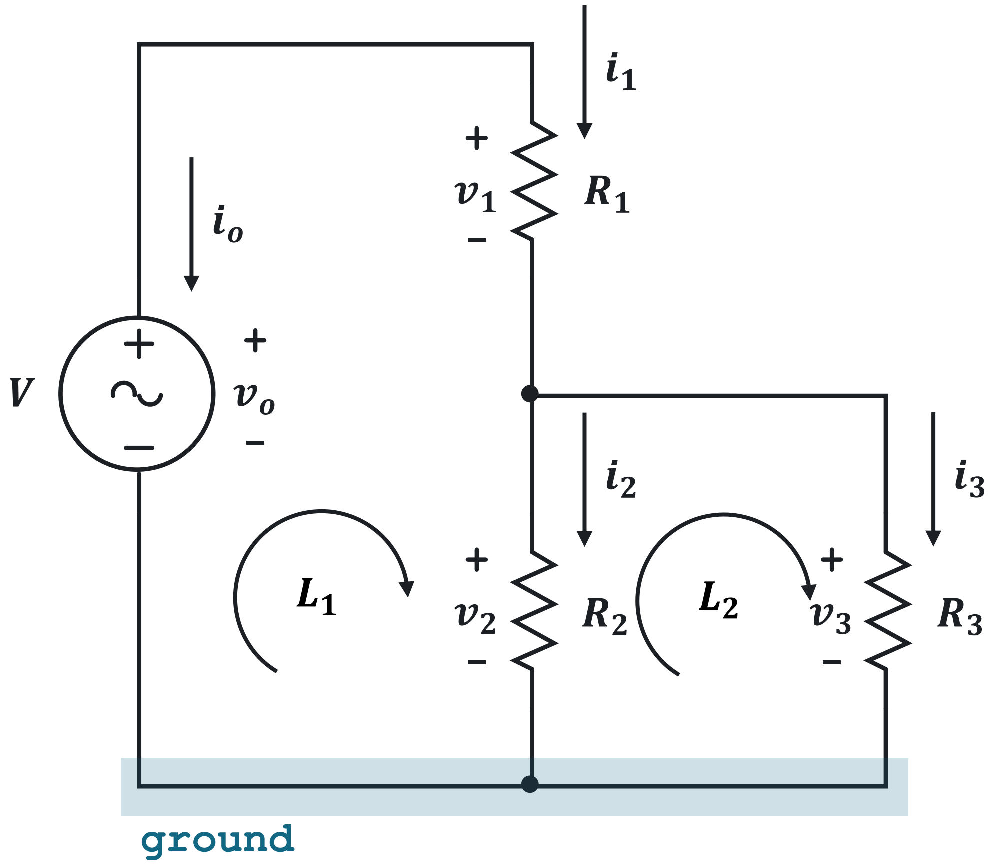
white – 9

Resistors have colored bands that represent specific digits, multipliers, and tolerance values.

LABELING VARIABLES



Steps in Labeling Variables

1. **Label the Reference Node (ground):**

Select a reference node with the most connections or the negative (−) terminal of a voltage source.

2. **Label Node Voltages:**

Mark higher potentials as positive (+) relative to the reference node.

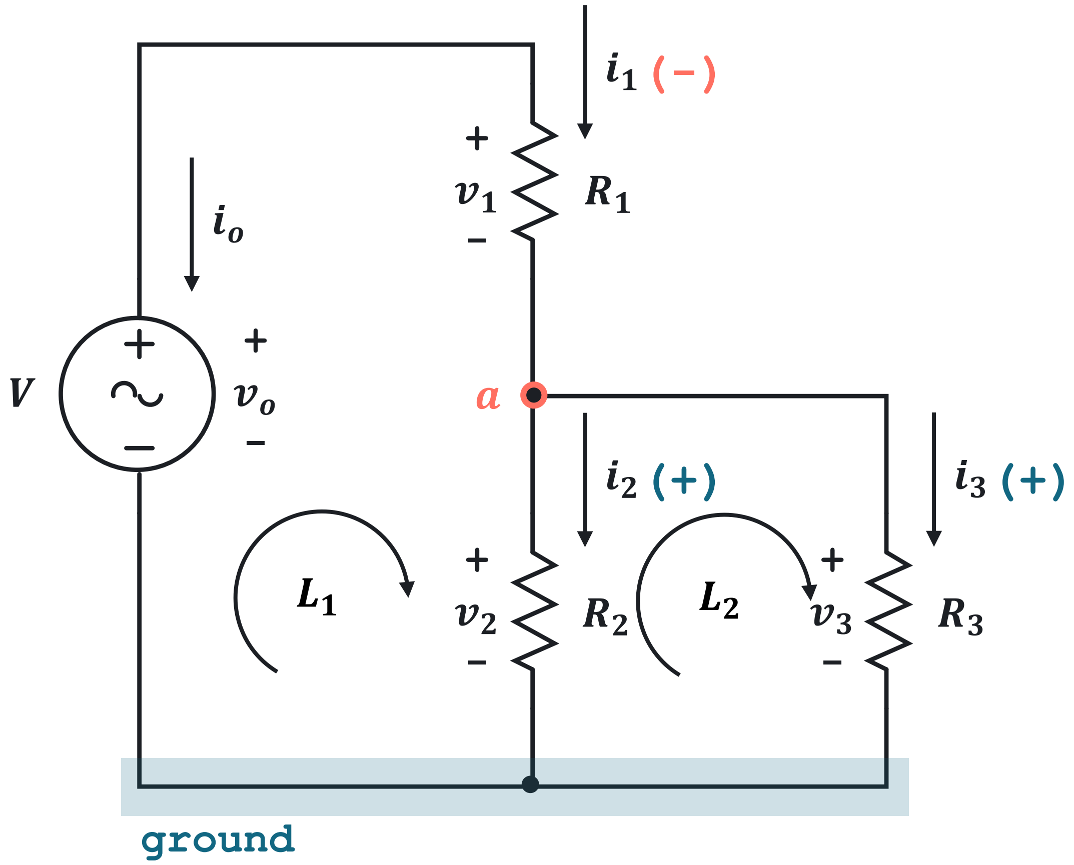
3. **Label Currents:**

Entering the positive (+) terminal of a component.

4. **Create a voltage loop:**

Follow the defined current directions.

CURRENT FLOW CONVENTION



Current Flow Convention

- Current **entering** a node is negative (–)
- Current **leaving** a node is positive (+)

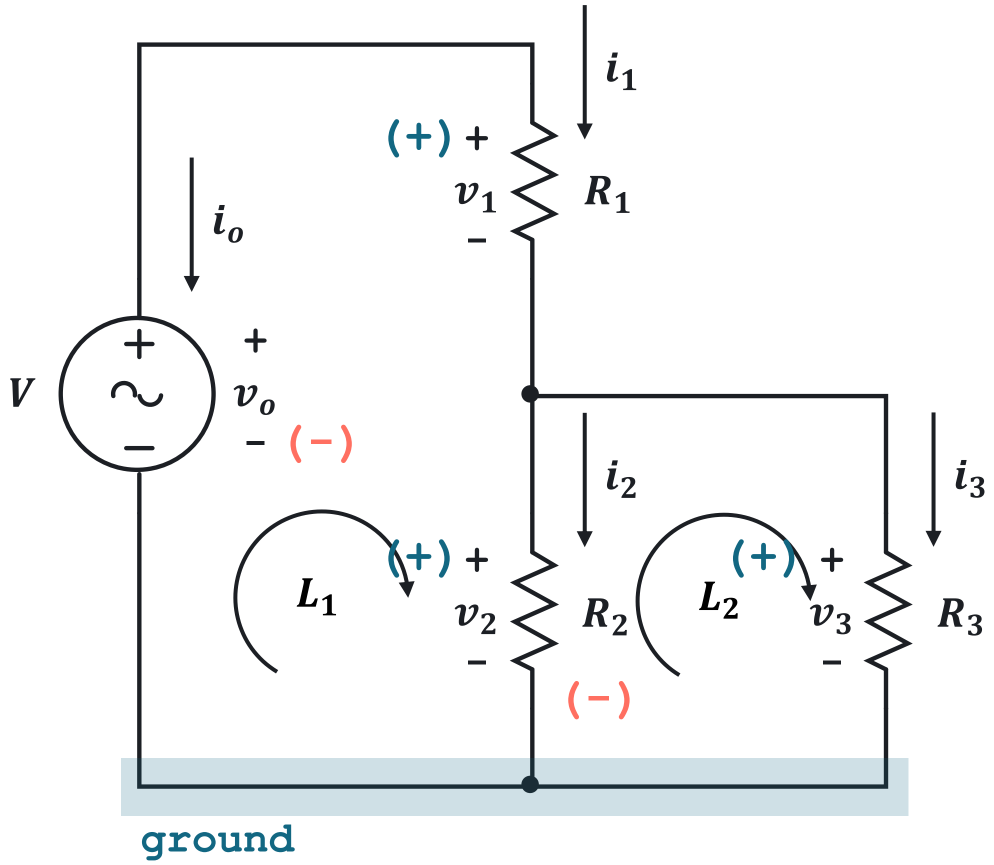
@*a*

$-i_1$

$+i_2$

$+i_3$

VOLTAGE LOOP CONVENTION



Voltage Loop Convention

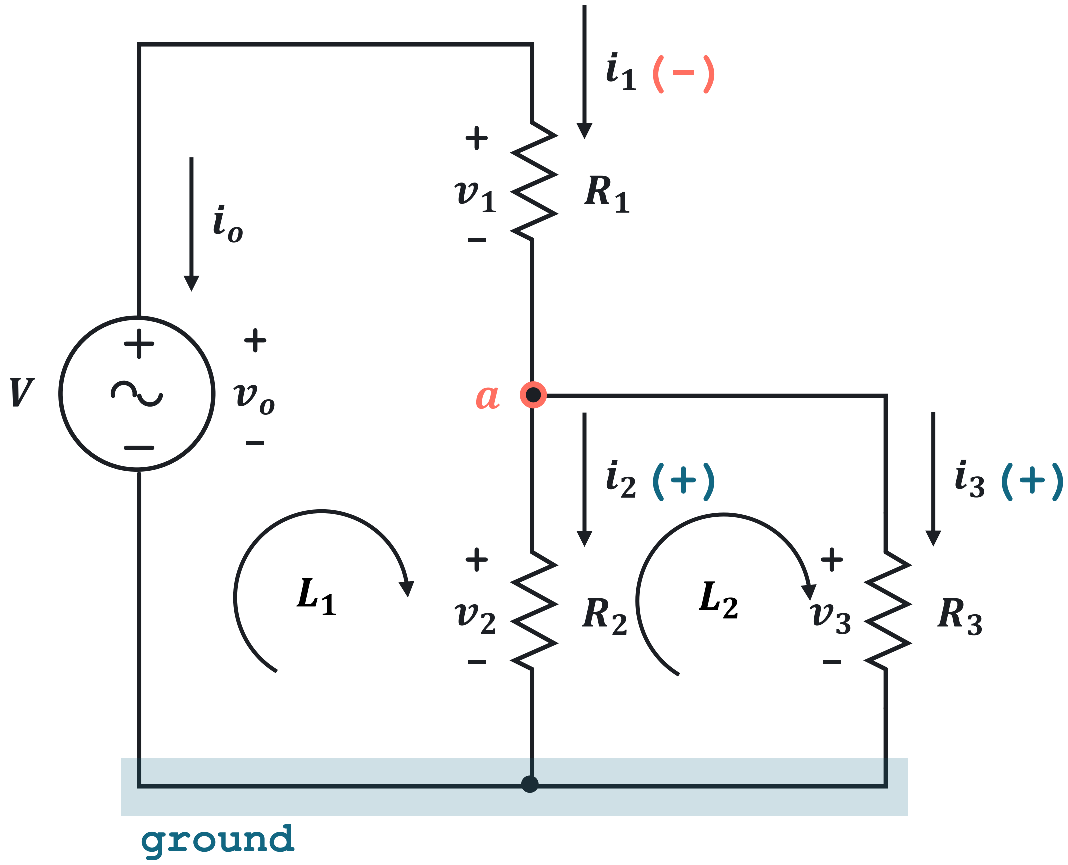
The “**sign**” of voltage of the element is the **first sign** the loop encounters.

$@L_1$	$@L_2$
$-v_o$	$-v_2$
$+v_1$	$+v_3$
$+v_2$	

KIRCHHOFF'S

CURRENT LAW AND VOLTAGE LAW

KCL



Kirchhoff's Current Law

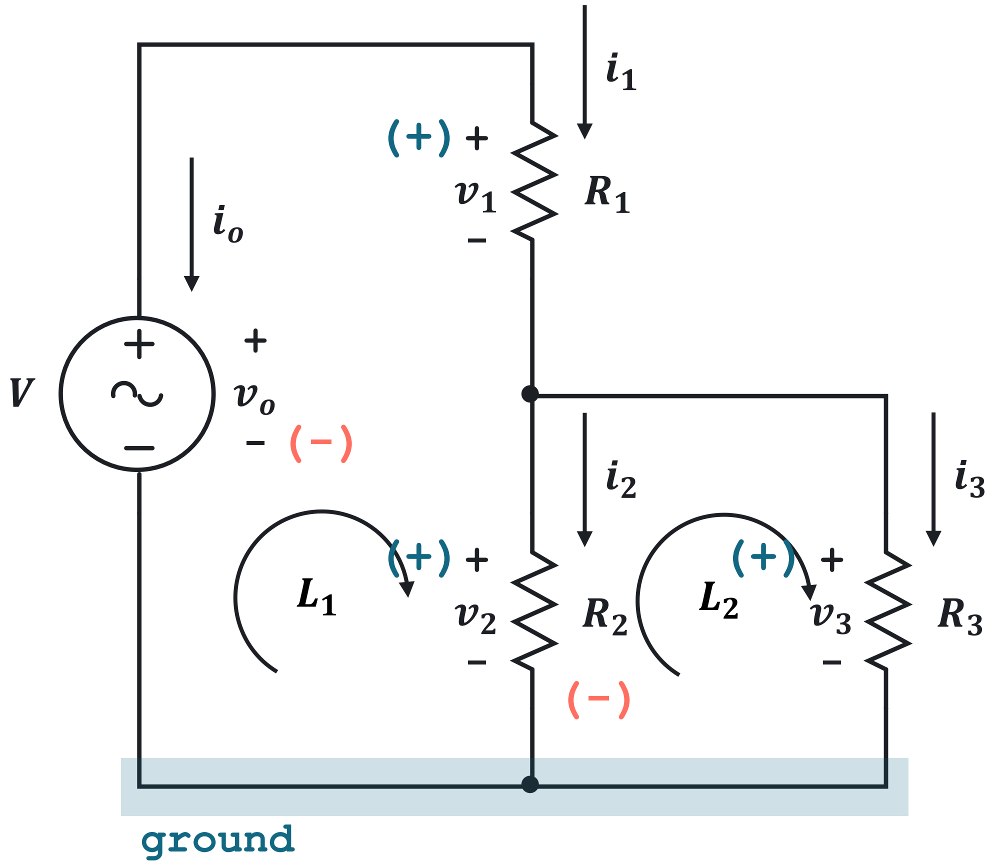
The summation of currents going-in and going-out a node is zero.

$$\sum i_j = 0$$

KCL @a

$$-i_1 + i_2 + i_3 = 0$$

KVL



Kirchhoff's Voltage Law

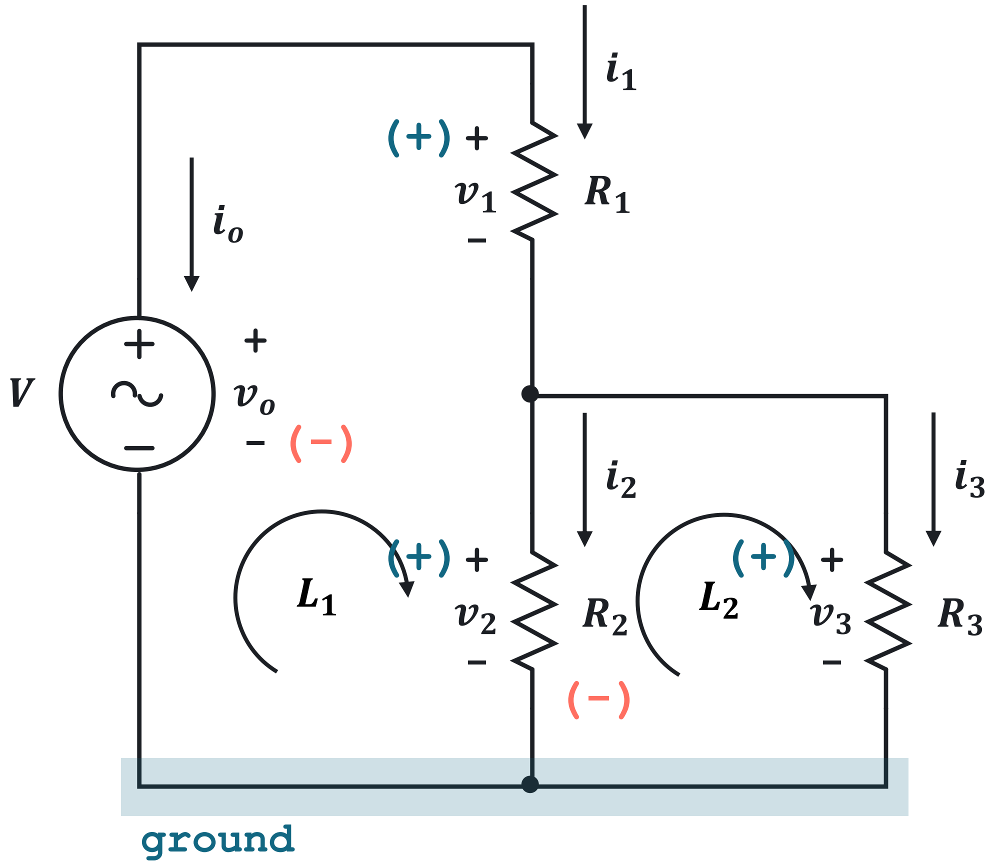
The summation of voltages in a closed-loop is zero.

$$\sum v_j = 0$$

KVL @L_1

$$-v_o + v_1 + v_2 = 0$$

KVL



Kirchhoff's Voltage Law

The summation of voltages in a closed-loop is zero.

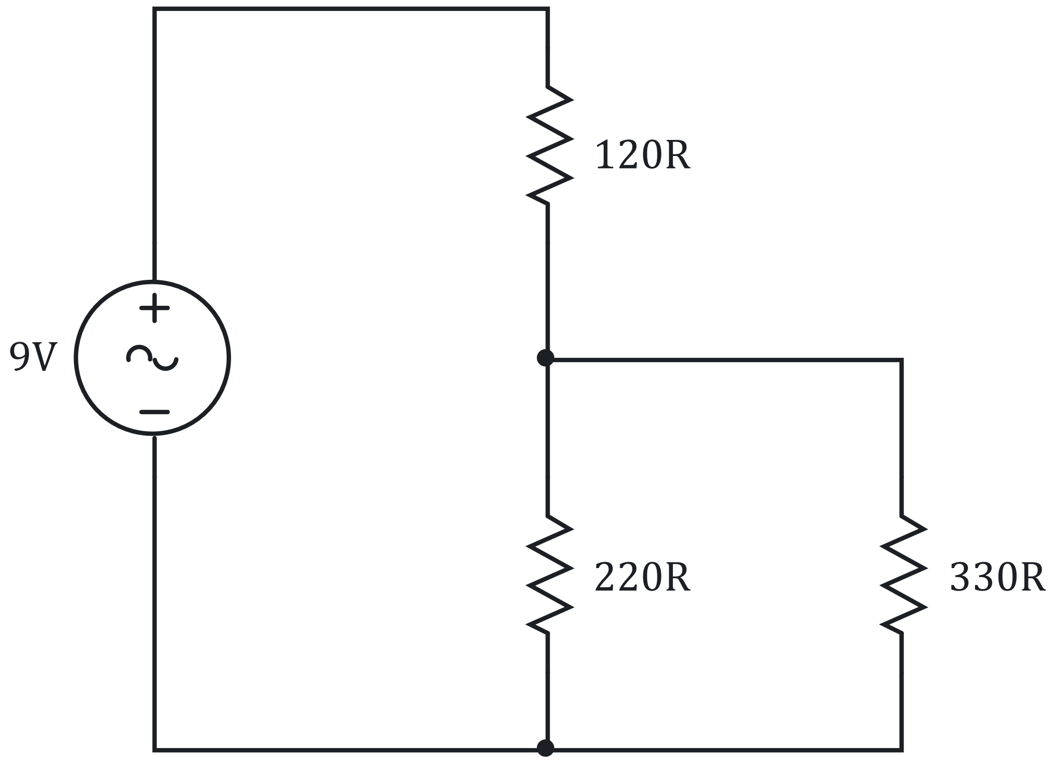
$$\sum v_j = 0$$

KVL @L_2

$$-v_2 + v_3 = 0$$

EXERCISE

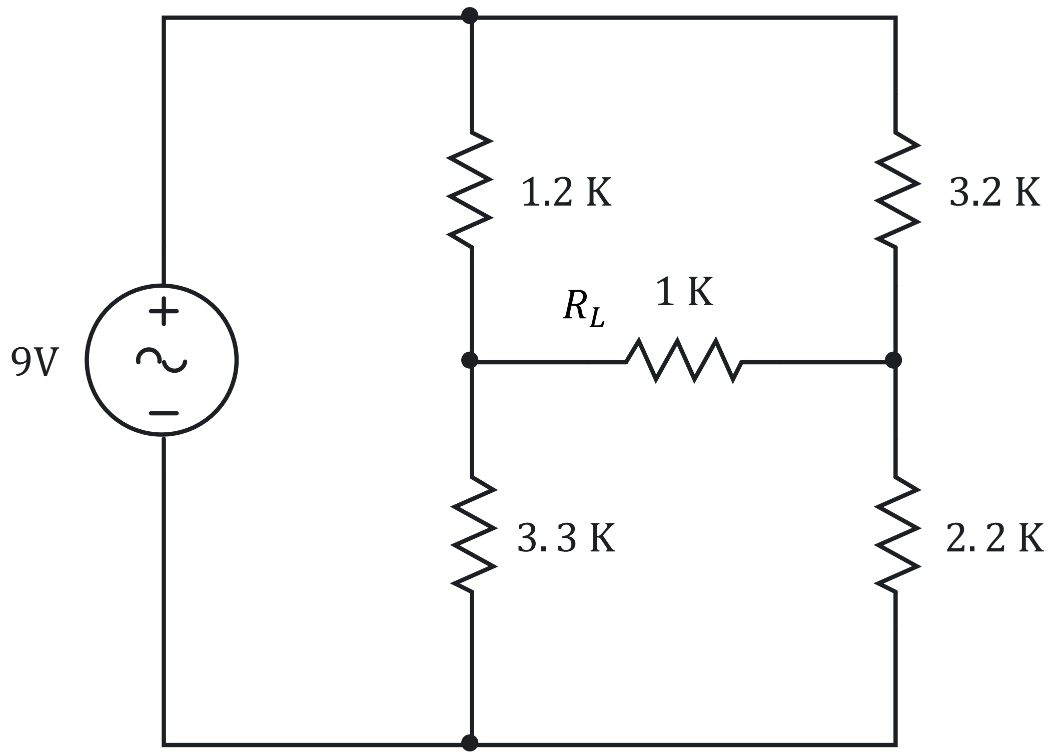
Determine the current flowing through each resistor and the voltage drop across each resistor in the given circuit.



Solution

EXERCISE

Determine the load R_L voltage and current of the given circuit.



Solution

LABORATORY